

AIRSTREET · REQUIREMENTS

AirStreet requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from AST-CAL-001; R15 redefined as the sensor-head cost (AST-DDR-001 D1); R6, R8 and R11 rest on adopted choices
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

AirStreet requirements

These requirements are checked by calculation in [AST-CAL-001](#). Targets remain proposals for review with the first co-design partner. R15 now covers the AirStreet sensor head only, with the FieldNode core costed in the FieldNode repo; this redefinition and the choices behind R6, R8 and R11 are adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review ([AST-DDR-001](#)). Three requirements are **not met** (R3, R13 and R15), six are at risk, three are met on paper and four are met by design.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (AST-CAL-001)
R1	Measure PM2.5 at street level	0 to 500 $\mu\text{g}/\text{m}^3$; hourly mean after humidity correction within $\pm 5 \mu\text{g}/\text{m}^3$ or $\pm 30 \%$ (whichever is larger) of a collocated reference	CalRig chamber check, then field collocation	At risk: within the bound up to 80 % RH; outside it above 85 % RH, where water uptake makes the correction uncertain (AST-CAL-001 E)
R2	Measure NO2 at street level	0 to 200 ppb (about 0 to 380 $\mu\text{g}/\text{m}^3$); hourly mean absolute error 5 ppb or less against a reference station after field calibration	Field collocation with a chemiluminescence reference station	At risk: hourly MAE about 3.9 ppb from a budget of literature and assumed terms (AST-CAL-001 C)
R3	Resolve NO2 near the WHO annual guideline	Annual mean uncertainty 2 $\mu\text{g}/\text{m}^3$ or less at 10 $\mu\text{g}/\text{m}^3$	Collocation statistics	Not met. Annual uncertainty about 4.2 $\mu\text{g}/\text{m}^3$ (1σ); a reference-to-street transfer bias does not average away (AST-CAL-001 C)
R4	Distinguish streets that exceed the EU NO2 limit value	Classify annual means above or below 40 $\mu\text{g}/\text{m}^3$ with 90 % confidence	Collocation statistics	At risk: decisive only for annual means below 34.6 or above 45.4 $\mu\text{g}/\text{m}^3$
R5	Measure temperature and humidity for corrections	$\pm 0.5 \text{ }^\circ\text{C}$ and $\pm 3 \%$ RH inside the shield; radiation error 1 $^\circ\text{C}$ or less in full sun at 1 m/s wind	CalRig chamber check; side-by-side field test against an aspirated reference	At risk: SHT45 meets the sensor figures; shield error 0.43 to 0.89 K at 1 m/s, which reads 4.2 % RH low at 85 % RH
R6	Calibrate with a stated method	T, RH and PM checked on CalRig before deployment; NO2 collocated with a reference station for at least 14 days before deployment and at least every 6 months	Calibration record per node	Met by design (AST-DDR-001 D3)
R7	Keep raw data for recalibration	Send raw NO2 working and auxiliary electrode signals, PM, T and RH with a calibration version tag	Firmware and data schema review	Met by design
R8	Report often enough for street maps	One record every 5 min; hourly means published within 15 min; 7 days of store and forward	Timing and storage calculation	Met on paper: 5 min records, hourly mean within about 5 min, 7 days in 40.3 kB
R9	Fit the FieldNode power budget	Sensor average 115 mW or less; energy neutral at the FieldNode design sun hours	Power budget calculation	Met on paper: 48.0 mW design load; 1.29 Wh/day drawn against 7.75 Wh stored in the worst month; 11.9 days without sun
R10	Stay within radio duty-cycle rules	LoRaWAN airtime below 1 % in EU868 sub-bands	Airtime calculation	Met on paper: 0.082 % at SF9, 0.60 % at SF12
R11			Design review	

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (AST-CAL-001)
	Mount on street poles without drilling	Poles 80 to 200 mm diameter; band clamps only; inlets between 1.5 and 4 m above ground		Met by design (inlet 3.0 m, AST-DDR-001 D7)
R12	Survive outdoors	Pod inlets face down behind insect mesh with drip lid; electronics in the FieldNode IP65 enclosure; -10 to 50 °C	Design review, later field trial	At risk: FieldNode is rated to 45 °C ambient; SPS30 recommended range 20 to 80 % RH; condensation and insects unverified
R13	Light and compact on the pole	Mass 3.5 kg or less; frontal area 0.12 m² or less	Massing model	Not met. 3.65 kg and 0.138 m² from the model; FieldNode is 2.41 kg, not the 1.7 kg assumed at TRL 2
R14	Collect levels only	No images, audio or personal identifiers collected or sent	Design review	Met by design
R15	Low cost	AirStreet sensor-head parts \$180 or less per node (<code>budget_usd</code>); FieldNode core excluded and costed in FieldNode	Priced BOM (AST-CAL-001 I)	Not met. Sensor head \$283.00, against \$180 (<code>budget_usd</code>) and the \$280 proposed in AST-DDR-001, awaiting Amish; \$409.00 with the FieldNode core
R16	Serviceable in the field	NO2 sensor and PM sensor replaceable in 15 min at the pole with hand tools	Design review	At risk: about 14 min estimated; a new NO2 sensor needs collocation, so a pre-collocated spare pod is the practical exchange unit

■ Assumptions

- NO2 conversions use 1 ppb = 1.88 µg/m³ at 25 °C and 101.325 kPa.
- The WHO guideline level for annual NO2 is 10 µg/m³ and for PM2.5 is 5 µg/m³, as cited by the European Environment Agency (EEA, 2024). The EU annual limit value for NO2 is 40 µg/m³ under Directive 2008/50/EC, which also sets sampling inlets between 1.5 m and 4 m above ground (EUR-Lex).
- FieldNode allows 115 mW average for sensors at its 5-day autonomy ceiling, with 100 mW proposed as its published design value (FND-CAL-001); AirStreet fits either.
- The \$280 sensor-head budget recommended in AST-DDR-001 D1 is recorded only; `budget_usd` in `project.yaml` stays \$180 until Amish decides.
- R1 and R2 targets are proposals for review, set with reference to published field results; they are not taken from a standard.